

Overview

In this lab you will write a program that implements a Java interface that describes some search-related methods. You will implement the class `ArraySearch` that implements the interface `Search`. Your implementation needs to work with an array of generic types which are `Comparable`.

Objectives

- Practice creating a class that implements an interface
- Creating and using non-static methods
- Perform comparisons of type `Comparable`
- Apply test cases to your program

Starter Code

Download the starter files from the course [public/19L](#) folder.

Tasks

- Implement all of the methods in the `Search` interface based on the comments in the interface.
- For determining if the arr is sorted, you may want to review the Java API for the [Comparable](#) interface. For example the following Integer arr, {2, 10, 12, 20, 20, 100}, is considered sorted since the elements are monotonically increasing.
- For the `index` method, you may use any search technique of your choosing. But remember if you try a binary search, it requires the list to be sorted.
- The array can contain zero elements, but if it contains any elements, each object is unique (meaning there are no duplicate values in the array passed to the constructor).